WESTERN QUANTUM CLUB

Quantum in Biology

Foundations + Case Studies

Today

- Quick primer on key quantum chemistry terms
- Case studies on molecule simulations
  - IBM
  - Google
  - Rigetti Computing
- Why these results matter for biology
  - Vision
  - Enzymes
  - Photosynthesis

Background

Terminology + Demos

Key Terms

Hamiltonian

Ansatz

Coherence

QPE

QFT

VQE (Variational Quantum Eigensolver)

- VQE is a hybrid quantum-classical algorithm used to estimate ground state energy
- The molecule’s Hamiltonian is mapped to qubits

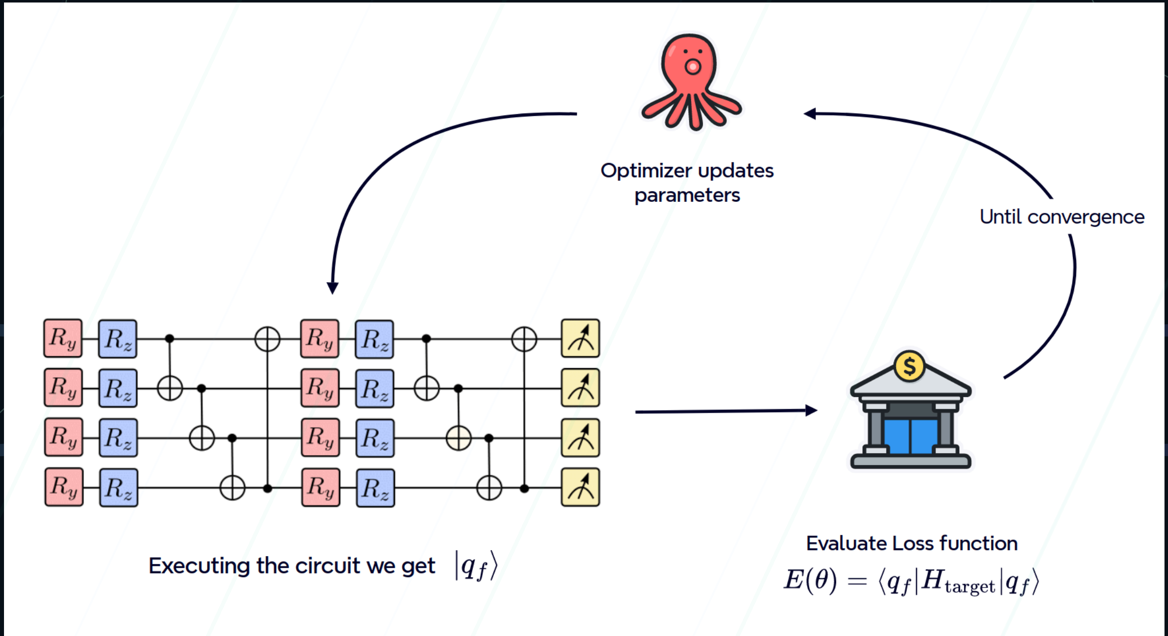


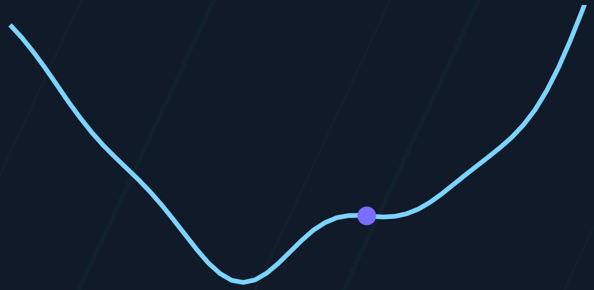
Figure: VQE loop. A parameterized circuit is optimized to minimize the measured energy.

Interactive: VQE Loop

VQE Loop Simulator

Parameter θ 1.60

Energy E(θ) 0.8298



Optimize Step | Reset

This is a simplified model. The “quantum” step is the energy measurement, the “classical” step updates θ to lower the energy.

Interactive: VQE vs QPE

VQE vs QPE

VQE | QPE

Pros

- Works on noisy (NISQ) hardware
- Shallow circuits
- Flexible ansatz choices

Cons

- Optimization can get stuck
- Accuracy depends on ansatz
- Many measurements needed

Circuit Depth

Shallow

Previous quantum studies in biology

Google: H_2 Energy Surface (2016)

- Quantum computers can estimate molecular ground state energies, which determine bonding, stability, and reactions[2]
- Demonstrates both VQE and phase estimation on a superconducting processor[2]
- This is a proof-of-concept step toward simulating complex biologcal molecules like proteins, enzymes, and drug compounds in the future[2]

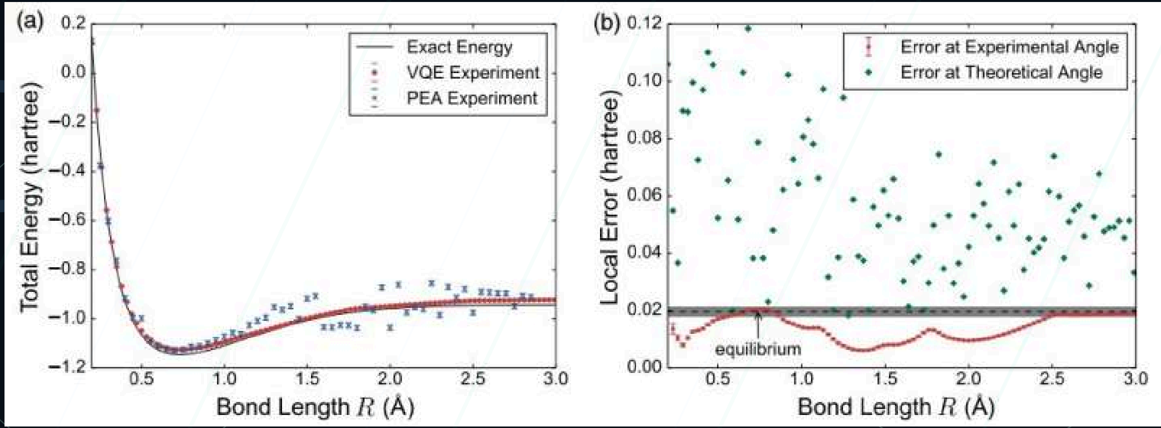


Figure: H_2 energy curve and errors (O'Malley et al., 2016).[4]

Interactive: H_2 Energy Curve

H_2 Bond-Length Energy Curve

Bond length r (Å) 0.90

Energy E(r) 0.1500

Minimum energy occurs near the equilibrium bond length

Interactive: Noise vs. Energy

Noise vs. Energy Estimation

Noise level 0.15

Measured energy 0.1725

Ideal Noisy

IBM: VQE on Small Molecules (2017)

- Useful for understanding how molcules form bonds and react[1]
- Knowing the ground state of mleucles is critic al in fields like drug discovery, materials science, and biochemistry[1]
- This is a proof-of-concept step toward simulating complex biologcal molecules like proteins, enzymes, and drug compounds in the future[1]

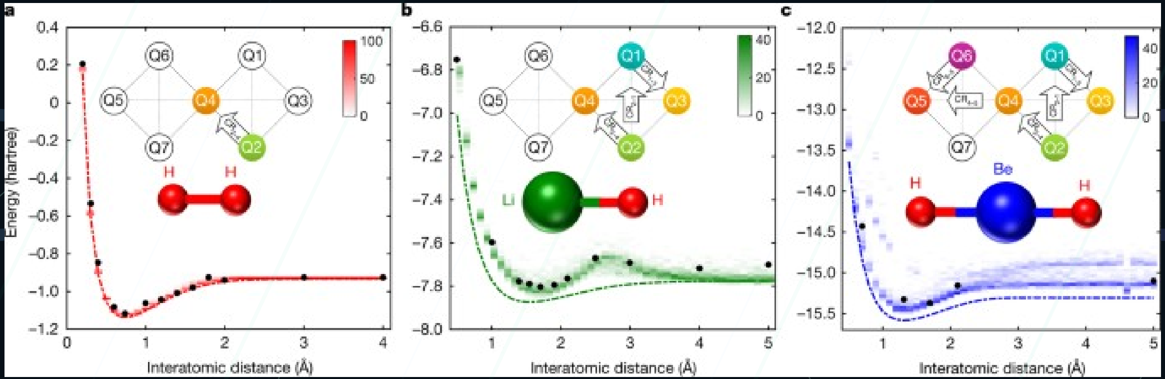
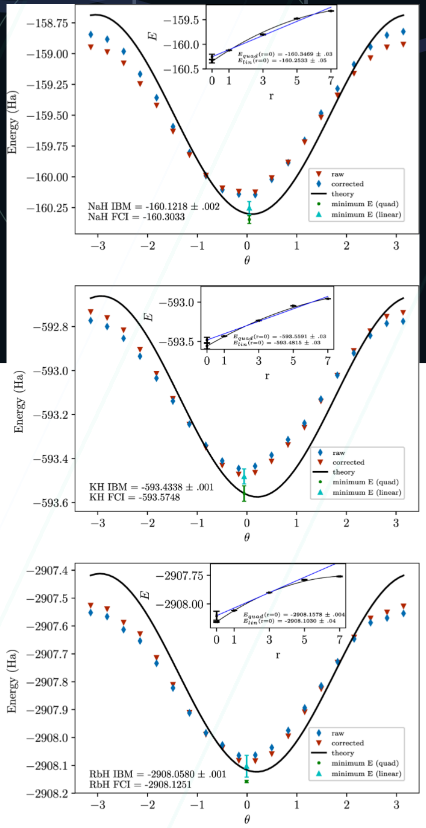


Figure: molecular energy curves for small molecules (Kandala et al., 2017).[1]

Rigetti Computing: Chemistry Benchmarking (2019)

- Demonstrated molecular energy estimation on a superconducting quantum processor[3]
- Tested chemically realistic molecules (NaH, KH, RbH)[3]
- Experimental results closely matches theoretical chemistry calculations[3]
- Showed that near-term quantum hardware can perform useful chemistry simulations[3]
- Important benchmark for scaling quantum chemistry algorithms[3]



Quantum in Nature

- Many biological processes depend on molecular structure and energy (bonding, stability, and reaction pathways)

Vision

Enzymes

Photosynthesis

- Today's demos (IBM/Google/Rigetti) are small, but they establish the workflow needed for larger biological systems

Resources

1. Kandala et al., *Nature* (2017) — VQE for small molecules. https://www.nature.com/articles/nature23879
2. O'Malley et al., *PRX* (2016) — Scalable molecular energies. https://research.google/pubs/scalable-quantum-simulation-of-molecular-energies/
3. McClean et al., *npj QI* (2019) — Chemistry benchmark. https://www.nature.com/articles/s41534-019-0209-0
4. Phys.org (2016) — Image credit for PRX figure. https://phys.org/news/2016-07-scalable-quantum-simulation-molecule.html
5. Skopintsev et al., *Nature* (2023) — Ultrafast vision. https://www.nature.com/articles/s41586-023-05863-6
6. Ranaghan et al., *Org. Biomol. Chem.* (2004) — Chorismate mutase QM/MM. https://doi.org/10.1039/B313759G
7. Engel et al., *Nature* (2007) — Photosynthetic coherence. https://www.nature.com/articles/nature05678